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Preventing Hand-Arm Vibration Syndrome in Industrial Workers: A Self-Powered Triboelectric Nanogenerator Glove for Real-Time Frequency Analysis and Hazard Detection

Introduction:

Hand-Arm Vibration, also known as HAVS, is a severe occupational condition and a secondary form of Raynaud's syndrome caused by "continuous use of vibrating hand-held machinery." [1]. Specifically, it is a vascular condition characterized by damage to circulatory vessels and nerves in parts of the body frequently exposed to hazardous vibrations, leading to muscle weakness and difficulty using hands due to numbness, among countless other symptoms [2]. Although there are certain treatment options which involve "reducing or stopping the use of vibrating power tools," making lifestyle changes to improve blood circulation, and prescription drugs to reduce the consequences of HAVS, "severe symptoms cannot always be reversed," leading to permanent damage as well as loss of function in the hands or fingers [2]. While exposure to vibrating tools in the frequency range of 8 Hz to 1500 Hz can be problematic, frequencies between 100 Hz and 150 Hz are considered to be the most dangerous in terms of developing HAVS [3]. Common tools associated with these frequencies and the development of related problems include "handheld drills, rotary saws, chainsaws, disc cutters, road rollers, concrete breakers, high-pressure hoses," and other power tools or heavy machinery, deeming tradespeople and industrial workers especially prone to developing HAVS [2]. Alarmingly, "about 25 % (approximately 2.5 billion)" of the global workforce is "exposed to occupational vibration", making Vibration-induced White Finger (VWF) the "most common vibration-related occupational disease" [4]. Furthermore, occupational diseases such as this are cumulative, meaning symptoms progress steadily over time, which can cause early warning signs to be ignored until significant damage has already occurred. When considered in conjunction, the prevalence of HAVS and its subtle persistence emphasize an urgency to find a suitable preventative tool to protect those constantly exposed to vibrational machinery in their line of work.

Literature Review:

Triboelectric devices have emerged as a promising foundation for developing wearable sensory tools for movements involved in physiological or physical processes. Triboelectricity involves capturing charges generated during contact between a material that acquires a positive charge upon frictional rubbing (tribo-positive) and one that obtains a negative charge (tribo-negative) using electrodes [5]. TENGs, or triboelectric nanogenerators, are engineering electronic devices that take advantage of triboelectricity at a nanometer scale. There are various methods to design and structure a TENG, with the most common form being the contact-separation mode, which involves two materials of distinct electron affinities that are separated by a gap coming into physical contact with each other under an external force, creating an electrostatic potential. This field drives electrons to be deposited on electrodes and then flow through an external energy-harvesting circuit [5]. Other approaches to fabrication, such as linear-sliding mode, which involves the triboelectric components sliding past one another, and single-electrode mode, where a single electrode and dielectric component periodically approach

and depart from each other, exist, but their application depends on the nature of the device in which the TENG is being used.

Though the triboelectric effect, responsible for the creation of static electricity, has been known to mankind for over two thousand years, novel ways of "capturing charges from the effect and harnessing electricity had not been seen before in the same way [5]. Notably, the first mention and in-depth investigation of TENGs appeared in a 2012 paper authored by Fan et al., which explored their potential in mobile and personal electronics [6]. Following this developmental revelation, an increasing number of devices have been utilizing triboelectric nanogenerators since they "offer high flexibility and energy density, making them more suitable for miniaturization and planar integration compared to piezoelectric and electrostatic harvesters" [7]. For instance, in 2014, Zheng and team developed a symbiotic cardiac pacemaker that relied on a TENG the size of a commercial coin battery to generate electricity from an individual's breathing. Remarkably, it was able to convert mechanical energy into electricity at 4.9 μW/cm3, enough " to charge a lithium-ion battery" [8]. The revolution marked a step towards developing a pacemaker that would not need to rely on batteries, potentially eliminating the need for surgical replacement after specific intervals of time. Similarly, recently in 2026, a device utilizing a foam built-in electrode triboelectric sensor system was developed for fall risk detection in construction workers [4].

Such advancements offer an optimistic perspective in the development of wearables to detect and prevent HAVS, a field where a scarcity of diverse preventative tools, mostly battery-reliant, undermines ongoing efforts to combat the prevalence of this occupational disease. Additionally, though TENGs continually prove to be an intriguing area of investigation and application, "extracting electrical energy from TENG remains challenging due to its time-varying nature and low internal capacitance" [7]. "[R]esearch on dedicated integrated power-conversion methods" is also currently limited, revealing a need to investigate the feasibility of TENG-based systems as self-powered detection tools [7].

Research Question & Aims:

How effective is a triboelectric nanogenerator (TENG)-based glove at detecting harmful vibrational frequencies associated with Hand-Arm Vibration Syndrome in construction workers?

Under this line of inquiry, this study intends to investigate:

1) Effectiveness of various triboelectric material combinations in terms of
    a) Charge generation
    b) Durability & environmental resistance
    c) Compatibility with textile integration
2) Overall performance of the glove through the lenses of:
    a) Sensitivity to frequencies within a hazardous range (100-150 Hz)
    b) Voltage and current output under varying vibrational intensities
    c) Accuracy of frequency detection
    d) Signal stability and repeatability
    e) Comfort (wearability)
3) Efficacy of the prototype's signal-processing module considering:

a) Analog-to-digital converter (ADC) sampling
b) Fast Fourier transform (FFT) extraction
c) Hazard classification
d) Signal amplification
e) Alert generation

Proposed Methodology: This is a quantitative, experimental approach.

1) Material Evaluation (Phase 1): The material evaluation phase aims to identify the most effective triboelectric material pair and orientation
    - Independent Variables: Material pair, orientation of pair (e.g., separation distance)
    - Dependent Variables: Open-circuit voltage output
    - Control Variables: Nominal contact area (same dimensions for all samples), triboelectric separation mode (contact-separation mode)
    - Procedure: 2-3 material pairs composed of differing components will each undergo controlled vibration using a shaker, with the voltage output being measured using an oscilloscope. To offer a holistic overview of the pairs, standardized loading tests will be used to evaluate tensile strength and elasticity. The distance between each component of the TENG will also be varied in order to test which material pair and orientation is ideal for current and voltage generation.

2) Frequency Sensitivity & Detection Accuracy (Phase 2): This stage focuses on whether the glove can detect hazardous vibration frequencies (100-150 Hz)
    - Independent Variable: Input vibration frequency (set by function generator & shaker)
    - Dependent variable: Frequency (detected via FFT), error compared to actual input
    - Control Variables: Material pair (single pair selected to be ideal), contact force, sampling rate/analysis window length
    - Procedure: Known frequencies are to be applied to the glove (25-250 Hz) using a shaker to observe whether the prototype can accurately detect and record the input.
3) Hazardous Classification & Alert System (Phase 3): This aspect aims to test the alert system/danger detection of the unit itself. Specifically, whether the microcontroller utilized can accurately do what it is intended to do.
    - Independent Variables: Vibration conditions (safe vs. hazardous)
    - Dependent Variables: Classification of frequency, alarm response (LED/buzzer)
    - Procedure: The prototype is to be exposed to controlled vibration conditions via a shaker and function generator. Upon detecting varying frequency outputs, the glove's accuracy in classifying them as either safe or hazardous will be analyzed as a percentage.
4) Data Collection & Analysis:
    - Multiple trials will be conducted for each condition the glove experiences, keeping the listed control variables constant. This will ensure repeatability and consistency of the results. Data logging capabilities are present in the equipment utilized, allowing it to measure data over time in order to produce continuous graphs which can be utilized for statistical analysis. Statistical comparison is to be

conducted on the material performance, detection accuracy (FFT vs input frequency), and the alert reliability (as a percentage).

5) Prototype Construction:
   - Layered triboelectric components arranged in a contact-separation mode configuration surrounded by two conductive electrodes on either side will make up the main energy-harvesting base of the glove. When it experiences mechanical vibration, repeated contact and separation between the layers generate voltage, which is routed through a passive full-bridge rectifier and smoothing capacitor in order to convert the alternating current into direct current. A microcontroller interprets signals and voltage produced by the TENG in order to analyze the frequencies.

Limitations/Drawbacks:

Since it is extremely unlikely for a TENG's positioning on testing equipment to be replicated in a completely identical manner for each trial, differences in alignment may cause variations among trials for voltage output. The effect of misalignment can be reduced using measurement markers or clamps to ensure consistency. Additionally, despite the nanogenerator's ability to produce high voltage, its usable current output is very small, which can limit the sensor's reliability. This roadblock can be overcome by utilizing signal amplifiers, which can improve the readability of the signals.

Project Logistics & Post-Program Plan:

The project is expected to span approximately 6 months, which are split into 3 distinct phases: material testing, hazard detection, and prototype refinement, which are described in further detail in the methodology. Funds will be used to purchase materials, specifically triboelectric bases, as well as microcontrollers and circuitry components for prototype development and testing. The experimental devices will be sourced from local facilities and labs. Following the completion of the project and experimentation, the next steps are likely publication in a relevant research journal or presentation at academic research symposiums.

Expected Outcomes & Conclusion:

Taken together, the planned methodology and research are anticipated to demonstrate the feasibility of triboelectric nanogenerators (TENGs) in acting as a preventative tool against Hand-Arm Vibration Syndrome (HAVS) while also producing a working prototype that can detect vibrations of a harmful frequency (100-150 Hz) and trigger real-time alerts. The expected outcome will not only serve as insight into triboelectric-based devices for practical use, but it will also serve as evidence that self-powered triboelectric wearables can help address the gaps of current battery-reliant technology, acting as an innovative alternative promoting protective safety.

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